Operating Systems
Week 13 Lecture 13

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Previous

- Thrashing concepts and activities

Today Agenda

- Overview of Mass Storage Structure
- Describe the physical structure of secondary storage devices and the effect of a device's structure on its uses
- Explain the performance characteristics of mass-storage devices

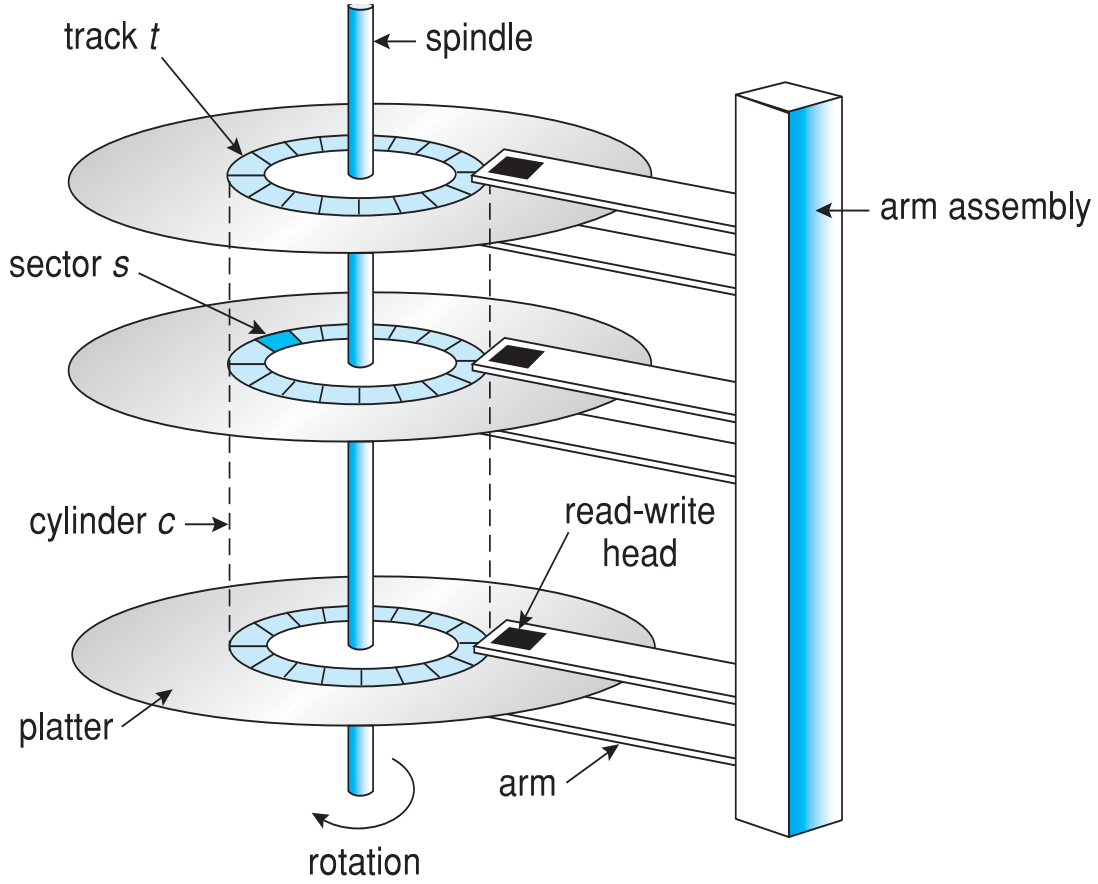
Mass Storage

Mass storage refers to devices that store **large amounts of data permanently**, even when the power is off.

Used to store the **operating system, software, and files**

- Provides **non-volatile** storage
- Examples: **Hard Disk Drive (HDD), Solid State Drive (SSD), USB flash drives**

Moving-head Disk Mechanism



Hard Disk Drives

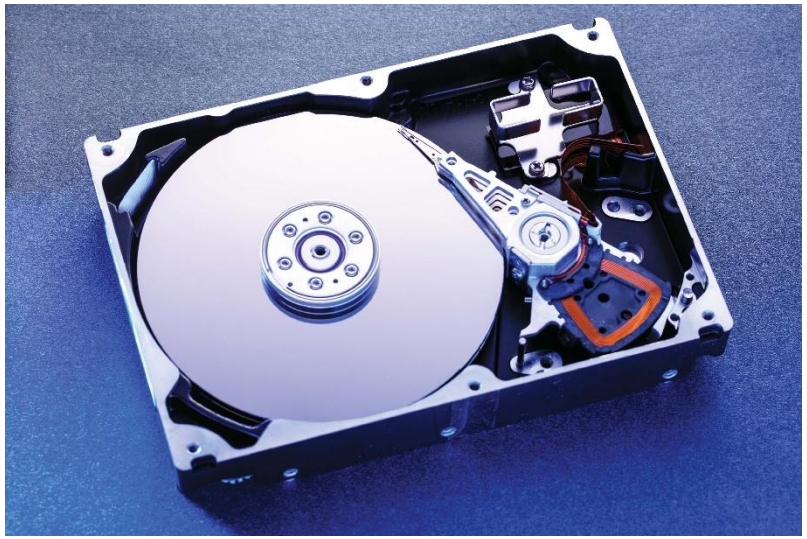
Platters range from .85” to 14” (historically)

- Commonly 3.5”, 2.5”, and 1.8”

Range from 30GB to 3TB per drive

Performance

- Transfer Rate – theoretical – 6 Gb/sec
- Effective Transfer Rate – real – 1Gb/sec
- Seek time from 3ms to 12ms – 9ms common for desktop drives
- Average seek time measured or calculated based on 1/3 of tracks
- Latency based on spindle speed
  - 1 / (RPM / 60) = 60 / RPM
- Average latency = ½ latency



Hard Disk Performance

Access Latency = **Average access time** = average seek time + average latency

- For fastest disk 3ms + 2ms = 5ms
- For slow disk 9ms + 5.56ms = 14.56ms

Average I/O time = average access time + (amount to transfer / transfer rate) + controller overhead

For example to transfer a 4KB block on a 7200 RPM disk with a 5ms average seek time, 1Gb/sec transfer rate with a .1ms controller overhead =

- 5ms + 4.17ms + 0.1ms + transfer time =
- Transfer time = 4KB / 1Gb/s * 8Gb / GB * 1GB / 1024^2KB = 32 / (1024^2) = 0.031 ms
- Average I/O time for 4KB block = 9.27ms + .031ms = 9.301ms

The First Commercial Disk Drive



1956
IBM RAMDAC computer included the IBM Model 350 disk storage system

5M (7 bit) characters
50 x 24” platters
Access time = < 1 second

Nonvolatile Memory

- Non-volatile memory **retains data even when power is turned off**.

- Stores the operating system, applications, and user data permanently.
- Used for booting the system and long-term data storage.

What happens when power is lost?

- Data remains safe.
- The OS can access saved files, applications, and system files after restarting.

Nonvolatile Memory Devices

- If disk-drive like, then called **solid-state disks** (**SSDs**)
- Other forms include **USB drives** (thumb drive, flash drive), DRAM disk replacements

- Can be more reliable than HDDs
- More expensive per MB
- Maybe have shorter life span – need careful management
- Less capacity
- much faster

Nonvolatile Memory Devices

Have characteristics that present challenges
Read and written in “page” increments (think sector) but can’t overwrite in place

- Must first be erased, and erases happen in larger ”block” increments
- Can only be erased a limited number of times before worn out – ~ 100,000
- Life span measured in **drive writes per day** (**DWPD**)
  - A 1TB NAND drive with rating of 5DWPD is expected to have 5TB per day written within warrantee period without failing



Magnetic Tape

Magnetic tape is a *non-volatile* memory.
It retains data even when power is turned off.
Data is stored using magnetic patterns on the tape surface.

Magnetic tape was used as an early secondary-storage medium. Although it is nonvolatile and can hold large quantities of data, its access time is slow compared with that of main memory and drives. In addition, random access to magnetic tape is about a thousand times slower than random access to HDDs and about a hundred thousand times slower than random access to SSDs so tapes are not very useful for secondary storage. Tapes are used mainly for backup, for storage of infrequently used information, and as a medium for transferring information from one system to another.

A tape is kept in a spool and is wound or rewound past a read–write head. Moving to the correct spot on a tape can take minutes, but once positioned, tape drives can read and write data at speeds comparable to HDDs. Tape capacities vary greatly, depending on the particular kind of tape drive, with current capacities exceeding several terabytes. Some tapes have built-in compression that can more than double the effective storage. Tapes and their drivers are usually categorized by width, including 4, 8, and 19 millimeters and 1/4 and 1/2 inch. Some are named according to technology, such as LTO-6 (Figure 11.5) and SDLT.



Figure 11.5 An LTO-6 Tape drive with tape cartridge inserted.

Non-volatile memory examples

Non-Volatile Memory	Description
HDD (Hard Disk Drive)	Traditional storage used to store OS, files, programs permanently.
SSD (Solid State Drive)	Faster, more modern non-volatile storage used by OS to boot quickly and load programs faster.
ROM (Read Only Memory)	Contains firmware and system bootloader. Used by OS during system startup (booting).
Flash Drives, Memory Cards	Used to store and transfer data; not part of OS but supported by it.

Non-Volatile Memory

Advantages of Non-Volatile Memory

- Retains data without power
- Crucial for system startup and file storage
- Portable (USB, SD cards)
- Becoming faster and more durable (SSD, Flash)

Limitations

- Generally **slower than volatile memory (RAM)** for real-time operations
- Limited **write endurance** (especially in flash memory)
- Can be **more expensive** per GB (especially SSDs)

Volatile Memory

DRAM frequently used as mass-storage device
- Not technically secondary storage because volatile, but can have file systems, be used like very fast secondary storage

RAM drives (with many names, including RAM disks) present as raw block devices, commonly file system formatted

Computers have buffering, caching via RAM, so why RAM drives?
- Caches / buffers allocated / managed by programmer, operating system, hardware
- RAM drives under user control
- Found in all major operating systems
  - Linux `/dev/ram`, macOS `diskutil` to create them, Linux `/tmp` of file system type `tmpfs`

Used as high speed temporary storage
- Programs could share bulk date, quickly, by reading/writing to RAM drive

Volatile memory

Volatile memory is memory that **requires power to retain data**.
When the power is turned off, **all data is lost**.

Used by the OS to store **currently running processes**, variables, buffers, and temporary data.

Enables **fast access** to data for CPU.

Volatile Memory is a type of memory that **requires continuous power** to store data.
When the power is turned off or the system crashes, **all data is lost**.

Structure Volatile Memory

Storage Mechanism
Volatile memory stores data using **electrical charges** in **capacitors or transistors**. The charge **must be refreshed constantly**.

Type	Storage Principle
DRAM (Dynamic RAM)	Stores each bit in a capacitor that leaks charge; must be refreshed
SRAM (Static RAM)	Stores each bit in flip-flop circuits using transistors; faster and more stable

Volatile Memory Types

Type	Description	Used in
RAM (Random Access Memory)	Temporarily holds OS and program data	Running apps, system tasks
DRAM	High-capacity, slower	Main memory in computers
SRAM	Faster, more expensive	CPU cache (L1, L2, L3)
Registers	Ultra-fast memory inside CPU	Instruction-level data
Cache Memory	Speeds up CPU access to frequent data	Between CPU and RAM

OS Implementation of Volatile memory

OS Role	Volatile Memory Use
Execution Space	OS loads itself and apps into RAM for execution
Process Management	Stores process data like variables, stack, heap
Buffers and Caches	Used for fast data transfer and storage
Temporary Storage	Data is discarded after shutdown or crash
Swapping/Paging	RAM is part of memory management for virtual memory

Volatile and non-volatile memory

Feature	Volatile Memory	Non-Volatile Memory
Power Dependency	Needs power to retain data	Keeps data even when power is off
Purpose	Temporary storage (for fast processing)	Long-term storage
Data Duration	Data is lost when system is turned off	Data stays until explicitly deleted
Speed	Very fast (RAM, Cache)	Slower (HDD, SSD)
Real-Life Example	Whiteboard you write on during class	Notebook you write notes in permanently
OS Examples	**RAM, CPU Cache**	**HDD, SSD, USB Drive, ROM**